ASHOKA WOMEN'S ENGINEERING COLLEGE

"SMART TRIAGE SYSTEM USING NLP AND ML"

BY:

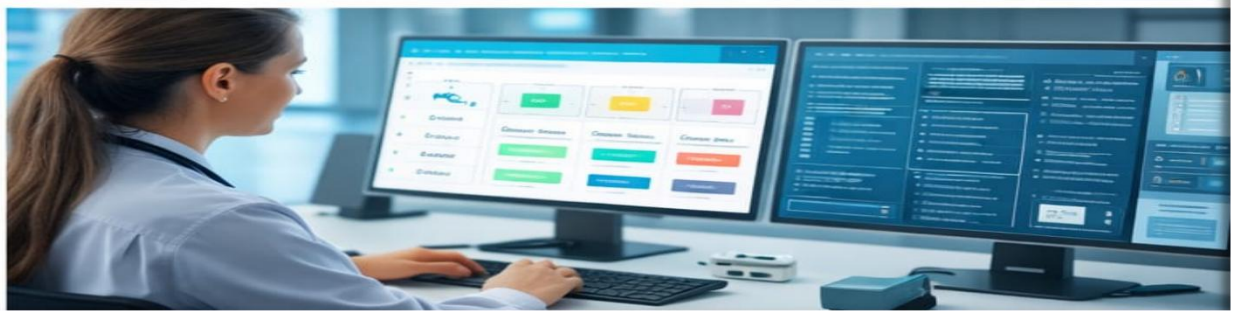
SHAIK AFREEN	222T1A3152
SHAIK SABIHA BANU	222T1A3153
SHAIK SUHANA	222T1A3154
SYED AAISHA	222T1A3157

Project Guide:
MR.M. CHINNA RAJU
Assistant Professor M.Tech(Phd), CSE

DEPT CSE-AI

ASHOKA
WOMEN'S ENGINEERING COLLEGE
(AUTONOMOUS)

"SMART TRIAGE SYSTEM USING NLP AND ML"



ASHOKA
WOMEN'S ENGINEERING COLLEGE
(AUTONOMOUS)

OUTLINES

- ABSTRACT
- INTRODUCTION
- EXISTING SYSTEM
- DISADVANTAGES OF EXISTING SYSTEM
- PROPOSED SYSTEM
- ADVANTAGES OF PROPOSED SYSTEM
- SYSTEM REQUIREMENTS(HARDWARE & SOFTWARE REQUIREMENTS)
- ALGORITHMS USED
- METHODOLOGY
- WORKING OF THE PROJECT(INTRODUCTION,CODE,RESULT)
- CONCLUSION
- FUTURE SCOPE
- APPLICATIONS
- REFERENCE

ASHOKA
WOMEN'S ENGINEERING COLLEGE
(AUTONOMOUS)

ABSTRACT

AI-Based Smart Triage Classification. The Smart Triage System uses NLP and ML to analyze patient symptom descriptions and classify cases into high, medium, or low urgency levels for efficient patient prioritization.Context-Aware and Adaptive Assessment.A fine-tuned BERT model understands clinical language and unstructured text, enabling accurate urgency prediction even with incomplete or varied symptom descriptions.Explainable Real-Time Decision Support .The system provides urgency predictions with probability scores and highlights key symptoms, offering transparent, real-time support to assist clinicians in timely decision-making.

ASHOKA
WOMEN'S ENGINEERING COLLEGE
(AUTONOMOUS)

INTRODUCTION

Challenges in Traditional Triage Systems. Healthcare systems rely on manual triage, which is time-consuming, inconsistent, and prone to human error, leading to delays in identifying high-risk patients and inefficient resource utilization. Need for Automated and Intelligent Triage. The increasing volume of patient data, especially in emergency departments and telemedicine, demands reliable AI-based solutions to ensure fast, accurate, and consistent patient prioritization. Role of AI in Modern Healthcare Triage
By leveraging NLP and ML techniques, automated triage systems can assist clinicians by improving decision accuracy, reducing workload, and enabling timely medical intervention.

ASHOKA
WOMEN'S ENGINEERING COLLEGE
(AUTONOMOUS)

EXISTING SYSTEM

- **Manual and Subjective Triage Process**
  Current triage relies heavily on clinician judgment and symptom checklists, making it time-consuming, inconsistent, and prone to delays during high patient volumes.
- **Rule-Based and Traditional ML Systems**
  Existing digital triage tools use predefined rules or basic ML models that lack contextual understanding and struggle with unstructured symptom descriptions.
- **Limited Explainability and Scalability**
  Most systems do not provide clear reasoning for triage decisions, reducing clinician trust and limiting scalability in modern healthcare environments.

ASHOKA
WOMEN'S ENGINEERING COLLEGE
(AUTONOMOUS)

DISADVANTAGES OF EXISTING SYSTEM

- **Heavy Dependence on Manual and Subjective Triage:**Reliance on human assessment leads to delays, fatigue-related errors, and inconsistent decisions, particularly in high-patient-volume environments.
- **Inability to Effectively Process Unstructured Symptom Data:**Rule-based and traditional ML systems struggle with free-text and ambiguous symptom descriptions, increasing the risk of misclassification and missed critical cases.
- **Lack of Contextual Understanding and Semantic Analysis:**Existing systems fail to capture relationships between multiple symptoms, resulting in inaccurate urgency prediction for complex or rare medical conditions.
- **Limited Explainability, Accuracy, and Scalability:**Inconsistent performance, extensive feature engineering, and lack of transparent reasoning reduce clinician trust and limit scalability in modern, data-intensive healthcare settings.

ASHOKA
WOMEN'S ENGINEERING COLLEGE
(AUTONOMOUS)

PROPOSED SYSTEM

- **NLP-Based Automated Symptom Analysis:**The proposed system uses advanced NLP techniques to interpret unstructured patient symptom descriptions, enabling accurate extraction of semantic and contextual information for reliable triage assessment.
- **BERT-Driven Context-Aware Classification:**A fine-tuned BERT model classifies patients into High, Medium, or Low urgency levels by capturing complex relationships and bidirectional context within symptom text.
- **Explainable and Confidence-Aware Decision Support:**The system provides confidence scores, keyword highlights, and human-readable explanations, allowing clinicians to understand, verify, and trust AI-generated triage recommendations.

ASHOKA
WOMEN'S ENGINEERING COLLEGE
(AUTONOMOUS)

ADVANTAGES OF PROPOSED SYSTEM

- **Fast, Accurate, and Consistent Triage:**Automates patient assessment using NLP and ML, enabling rapid and consistent urgency classification while reducing delays and human variability.

- **Advanced Understanding of Unstructured Symptom Data:**BERT-based models accurately interpret complex, incomplete, or informal patient descriptions, improving reliability compared to traditional systems.

- **Scalable and Workflow-Friendly Deployment:**Supports real-time assessment, multiple concurrent users, and seamless integration with clinical and telemedicine platforms, improving efficiency and patient outcomes.

SYSTEM REQUIREMENTS(HARDWARE AND SOFTWARE REQUIREMENTS)

Hardware Requirements:

- Processor : Intel i5
- RAM : 8GB
- Storage : 256GB

Software Requirements:

- OS : Window 11
- python : 3.10
- Anaconda

ASHOKA
WOMEN'S ENGINEERING COLLEGE
(AUTONOMOUS)

ALGORITHMS USED

ALGORITHM USED:

To enable accurate and automated patient triage, a Natural Language Processing (NLP) and Machine Learning–based approach is adopted using a BERT-based Transformer model. The algorithm processes unstructured patient symptom descriptions, captures contextual and semantic relationships, and classifies cases into High, Medium, or Low urgency levels with confidence and explainability.

BERT (Bidirectional Encoder Representations from Transformers):

- **Purpose:** Used to classify patient symptom descriptions into triage levels based on urgency.
- **Working:** Processes symptom text bidirectionally through multiple transformer layers to capture context and relationships between words.
- **Text Processing:** Uses a pre-trained BERT tokenizer to convert symptom descriptions into subword tokens and contextual embeddings.
- **Classification:**Applies a softmax layer to generate probabilities for High, Medium, and Low triage classes.
- **Interpretability:**Supports keyword highlighting and confidence scoring to explain predictions to clinicians.
- **Advantages:**High accuracy with unstructured text, contextual understanding, and transparent decision support.

ALGORITHMS USED

Logistic Regression:

- **Purpose:**Used for probabilistic triage classification and comparison with advanced models.
- **Working:**Estimates the probability of each triage class using a logistic function based on input features.
- **Feature Handling:**Operates on structured or vectorized symptom data.
- **Interpretability:**Highly interpretable due to clear feature-weight relationships.
- **Advantages:**Simple, fast, and provides reliable probability estimates.

ASHOKA
WOMEN'S ENGINEERING COLLEGE
(AUTONOMOUS)

METHODOLOGY

Introduction:

The Smart Triage System uses Natural Language Processing (NLP) and Machine Learning (ML) to analyze patient-reported symptom descriptions and predict medical urgency levels. It assists healthcare providers in prioritizing patients efficiently, ensuring timely care for critical cases.

Key Points:

- **Data Collection:**Patient symptom descriptions along with corresponding triage labels (High, Medium, Low) are collected from clinical and simulated datasets to support supervised learning.
- **Data Preprocessing:**The collected text data is cleaned, tokenized, padded or truncated, and label encoding is applied to convert categorical triage levels into numerical form suitable for model training.

Model Selection and Training:

- **Supervised Learning:**A pre-trained BERT-based Transformer model is fine-tuned to classify patient symptoms into High, Medium, and Low urgency levels using contextual embeddings.

ASHOKA
WOMEN'S ENGINEERING COLLEGE
(AUTONOMOUS)

METHODOLOGY

- **Text Embedding**:BERT tokenizer and embeddings are used to capture semantic and contextual relationships within unstructured symptom descriptions.

- **Model Evaluation:**The trained model is evaluated using accuracy, F1-score, and confusion matrix to ensure reliable and generalized performance.

- **Deployment:**The system is deployed as a Flask-based web application to provide real-time triage predictions with confidence scores and explanations.

WORKING OF THE PROJECT

PREDICTED MODEL

```
import os
import torch
from transformers import BertTokenizer, BertForSequenceClassification
from sklearn.preprocessing import LabelEncoder
import pandas as pd

MODEL_DIR = "triage_bert_model"
TOKENIZER_DIR = "triage_bert_tokenizer"
DATASET_PATH = "dataset.csv"
MAX_LEN = 128

device = torch.device("cuda" if torch.cuda.is_available() else "cpu")

tokenizer = BertTokenizer.from_pretrained(TOKENIZER_DIR)
model = BertForSequenceClassification.from_pretrained(MODEL_DIR)
model.to(device)
model.eval()

if not os.path.exists(DATASET_PATH):
    raise FileNotFoundError(
        f"Dataset file not found at '{DATASET_PATH}'. This script rebuilds label mapping from the dataset. "
        "If you don't have the dataset, either save label_encoder during training or use the fallback script."
    )

df = pd.read_csv(DATASET_PATH)
if 'triage_level' not in df.columns:
    raise ValueError("Dataset does not contain 'triage_level' column required to rebuild label mapping.")

le = LabelEncoder()
```

ASHOKA
WOMEN'S ENGINEERING COLLEGE
(AUTONOMOUS)

WORKING OF THE PROJECT

```python
le = LabelEncoder()
le.fit(df['triage_level'].astype(str))  # ensure strings

def predict_triage(text: str) -> str:
    encoding = tokenizer.encode_plus(
        text,
        add_special_tokens=True,
        max_length=MAX_LEN,
        padding="max_length",
        truncation=True,
        return_attention_mask=True,
        return_tensors="pt"
    )
    input_ids = encoding["input_ids"].to(device)
    attention_mask = encoding["attention_mask"].to(device)

    with torch.no_grad():
        outputs = model(input_ids=input_ids, attention_mask=attention_mask)
        logits = outputs.logits
        pred_id = int(torch.argmax(logits, dim=-1).cpu().numpy()[0])

    label = le.inverse_transform([pred_id])[0]
    return label

if __name__ == "__main__":
    text = input("Enter text to classify: ").strip()
    if not text:
        print("No input provided.")
    else:
```

ASHOKA
WOMEN'S ENGINEERING COLLEGE
(AUTONOMOUS)

WORKING OF THE PROJECT

```
try:
    pred = predict_triage(text)
    print("\nPredicted TRIAGE LEVEL:", pred)
except Exception as e:
    print("Prediction failed:", e)
```

WORKING OF THE PROJECT

TRAINING MODEL

```
import pandas as pd
import ast
import torch
from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split
from torch.utils.data import Dataset, DataLoader
from transformers import BertTokenizer, BertForSequenceClassification
from torch.optim import AdamW
from transformers import get_linear_schedule_with_warmup
from tqdm import tqdm

device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')

df = pd.read_csv("datasetall.csv")

def join_tokens(token_list_str):
    try:
        tokens = ast.literal_eval(token_list_str)
        if isinstance(tokens, list):
            return " ".join(tokens)
        return ""
    except:
        return ""

df['text'] = df['processed_text'].apply(join_tokens)

le = LabelEncoder()
df['triage_label'] = le.fit_transform(df['triage_level'])
```

ASHOKA
WOMEN'S ENGINEERING COLLEGE
(AUTONOMOUS)

WORKING OF THE PROJECT

```
X_train, X_val, y_train, y_val = train_test_split(
    df['text'], df['triage_label'],
    test_size=0.2, random_state=42, stratify=df['triage_label']
)

class TriageDataset(Dataset):
    def __init__(self, texts, labels, tokenizer, max_len=128):
        self.texts = texts
        self.labels = labels
        self.tokenizer = tokenizer
        self.max_len = max_len

    def __len__(self):
        return len(self.texts)

    def __getitem__(self, idx):
        text = str(self.texts.iloc[idx])
        label = self.labels.iloc[idx]
        encoding = self.tokenizer.encode_plus(
            text,
            add_special_tokens=True,
            max_length=self.max_len,
            padding='max_length',
            truncation=True,
            return_attention_mask=True,
            return_tensors='pt'
        )
        return {
            'input_ids': encoding['input_ids'].flatten(),
```

ASHOKA
WOMEN'S ENGINEERING COLLEGE
(AUTONOMOUS)

WORKING OF THE PROJECT

```
            'attention_mask': encoding['attention_mask'].flatten(),
            'labels': torch.tensor(label, dtype=torch.long)
        }

tokenizer = BertTokenizer.from_pretrained('distilbert-base-uncased')

train_dataset = TriageDataset(X_train, y_train, tokenizer)
val_dataset = TriageDataset(X_val, y_val, tokenizer)

train_loader = DataLoader(train_dataset, batch_size=16, shuffle=True)
val_loader = DataLoader(val_dataset, batch_size=16)

model = BertForSequenceClassification.from_pretrained(
    'distilbert-base-uncased',
    num_labels=4
)
model.to(device)

optimizer = AdamW(model.parameters(), lr=2e-5)
total_steps = len(train_loader) * 3  # epochs
scheduler = get_linear_schedule_with_warmup(optimizer, num_warmup_steps=0, num_training_steps=total_steps)

epochs = 3
for epoch in range(epochs):
    model.train()
    loop = tqdm(train_loader, leave=True)
    for batch in loop:
        optimizer.zero_grad()
        input_ids = batch['input_ids'].to(device)
        attention_mask = batch['attention_mask'].to(device)
```

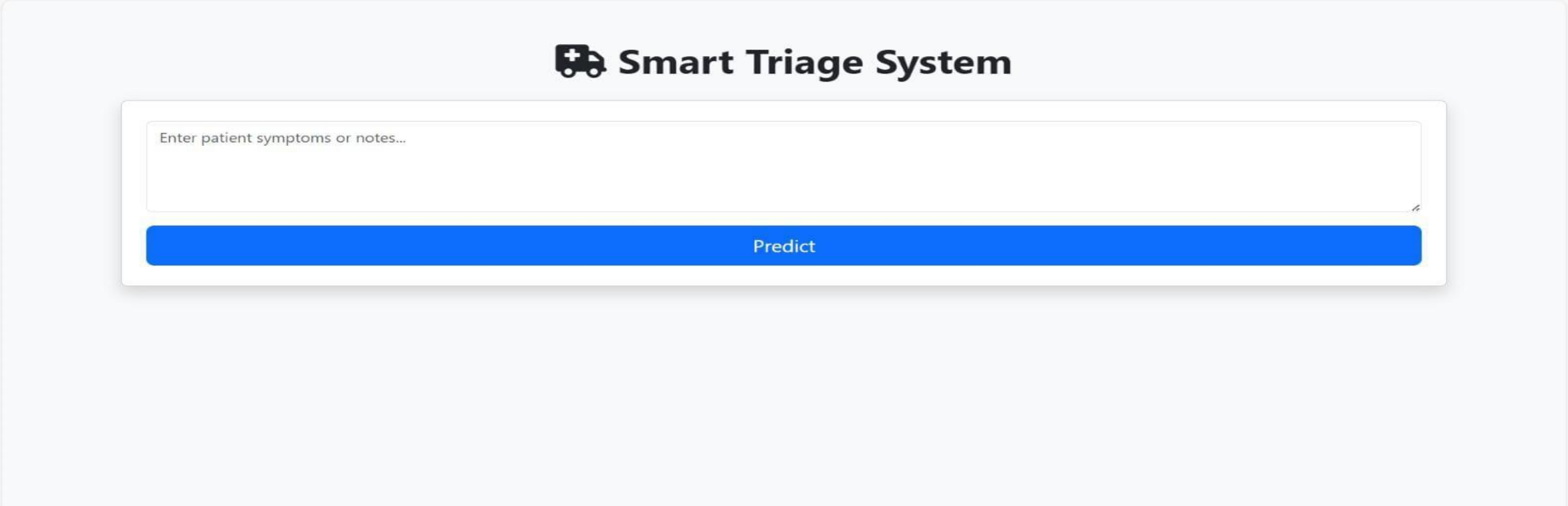
ASHOKA
WOMEN'S ENGINEERING COLLEGE
(AUTONOMOUS)

WORKING OF THE PROJECT

```
    num_labels=4
)
model.to(device)

optimizer = AdamW(model.parameters(), lr=2e-5)
total_steps = len(train_loader) * 3  # epochs
scheduler = get_linear_schedule_with_warmup(optimizer, num_warmup_steps=0, num_training_steps=total_steps)

epochs = 3
for epoch in range(epochs):
    model.train()
    loop = tqdm(train_loader, leave=True)
    for batch in loop:
        optimizer.zero_grad()
        input_ids = batch['input_ids'].to(device)
        attention_mask = batch['attention_mask'].to(device)
        labels = batch['labels'].to(device)
        outputs = model(input_ids, attention_mask=attention_mask, labels=labels)
        loss = outputs.loss
        loss.backward()
        optimizer.step()
        scheduler.step()
        loop.set_description(f"Epoch {epoch+1}")
        loop.set_postfix(loss=loss.item())

model.save_pretrained("triage_bert_model")
tokenizer.save_pretrained("triage_bert_tokenizer")
print("BERT training complete. Model saved.")
```

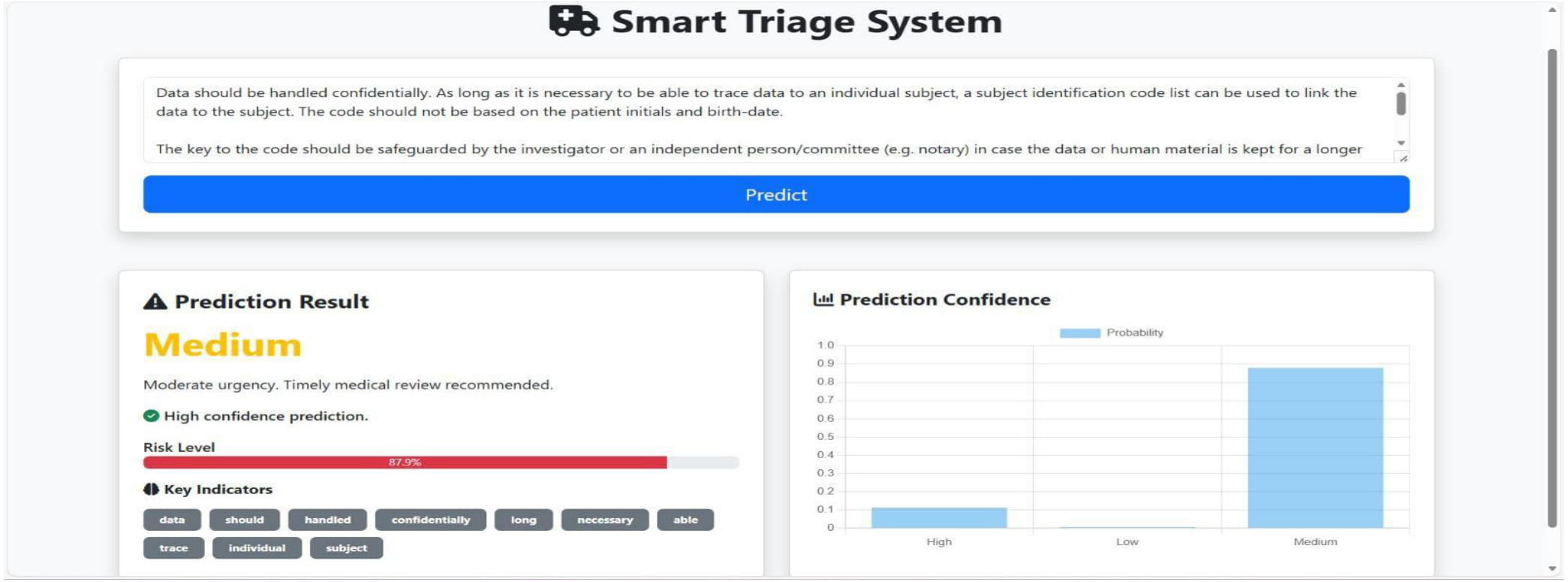
ASHOKA
WOMEN'S ENGINEERING COLLEGE
(AUTONOMOUS)

OUTPUT



ASHOKA
WOMEN'S ENGINEERING COLLEGE
(AUTONOMOUS)

WORKING OF THE PROJECT



ASHOKA
WOMEN'S ENGINEERING COLLEGE
(AUTONOMOUS)

CONCLUSION

- The system delivers accurate automated triage by using a BERT-based NLP model to understand complex and unstructured symptom descriptions, ensuring reliable prioritization while reducing clinician workload and human error.
- Transparency is achieved through confidence scores, keyword highlighting, and textual explanations, enabling clinicians to trust, verify, and effectively adopt AI-assisted triage decisions in real healthcare settings.
- The web-based Flask deployment enables real-time, remote access and supports multiple users simultaneously, making the system practical for telemedicine, emergency care, and primary healthcare environments.
- By accurately identifying high-risk patients and supporting efficient resource allocation, the system improves clinical efficiency, speeds up medical intervention, and contributes to better overall patient outcomes.

ASHOKA
WOMEN'S ENGINEERING COLLEGE
(AUTONOMOUS)

FUTURE SCOPE

- The system can be expanded with multilingual NLP capabilities and mobile application deployment, improving accessibility and inclusiveness for diverse populations and enabling seamless remote triage through smartphones.
- Integration with wearable devices and electronic health records (EHR) can enhance triage accuracy by combining symptom text with real-time physiological data while streamlining clinical workflows.
- Continuous and online learning mechanisms can allow the model to adapt to new diseases, evolving symptoms, and emerging healthcare trends, ensuring long-term reliability and relevance.
- Advanced explainability, predictive analytics, and integration with clinical decision support systems can enable deeper clinical insights, early risk prediction, and large-scale population health monitoring for proactive healthcare planning.

ASHOKA
WOMEN'S ENGINEERING COLLEGE
(AUTONOMOUS)

APPLICATIONS

- The Smart Triage System can be effectively applied in emergency departments to rapidly prioritize patients, ensuring critical cases receive immediate attention in high-pressure environments. In telemedicine and primary care settings, it supports remote symptom assessment, helping clinicians manage consultations efficiently and identify patients requiring urgent follow-up. The system is also valuable in ambulance dispatch and hospital admission screening, where early triage improves preparedness, reduces waiting times, and streamlines patient flow. When integrated with health apps or wearable devices, it enables remote health monitoring and timely alerts for concerning symptoms. In disaster and mass-casualty scenarios, the system assists in handling large patient volumes by optimizing triage and resource allocation. Additionally, it serves as a clinical decision support and research tool, contributing to medical training, public health monitoring, and data-driven healthcare improvements.

ASHOKA
WOMEN'S ENGINEERING COLLEGE
(AUTONOMOUS)

APPLICATIONS

- Beyond clinical use, the Smart Triage System can support health insurance screening by providing preliminary urgency assessments that help prioritize claims and authorize timely coverage for high-risk cases. It can also be deployed as a training and educational tool for medical students and newly appointed healthcare staff, allowing them to practice triage decision-making using realistic symptom inputs. On a broader scale, anonymized data generated by the system can be used for public health surveillance, helping authorities identify disease trends, outbreak clusters, and regional healthcare demands. By bridging patient-reported data with intelligent analysis, the system contributes to more proactive, data-driven, and resilient healthcare ecosystems.

ASHOKA
WOMEN'S ENGINEERING COLLEGE
(AUTONOMOUS)

REFERENCES

1. Devlin, J., Chang, M. W., Lee, K., & Toutanova, K. (2019).
   BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding.
   Proceedings of NAACL-HLT 2019, pp. 4171–4186.
2. Vaswani, A., Shazeer, N., Parmar, N., et al. (2017).
   Attention Is All You Need.
   Advances in Neural Information Processing Systems (NeurIPS 2017).
3. Rajkomar, A., Dean, J., & Kohane, I. (2019).
   Machine Learning in Medicine.
   New England Journal of Medicine, 380(14), 1347–1358.
4. Esteva, A., Robicquet, A., Ramsundar, B., et al. (2019).
   A guide to deep learning in healthcare.
   Nature Medicine, 25, 24–29.

ASHOKA

WOMEN'S ENGINEERING COLLEGE

(AUTONOMOUS)

THANK YOU